Speech Synthesis

Nagaraj Adiga

Outline

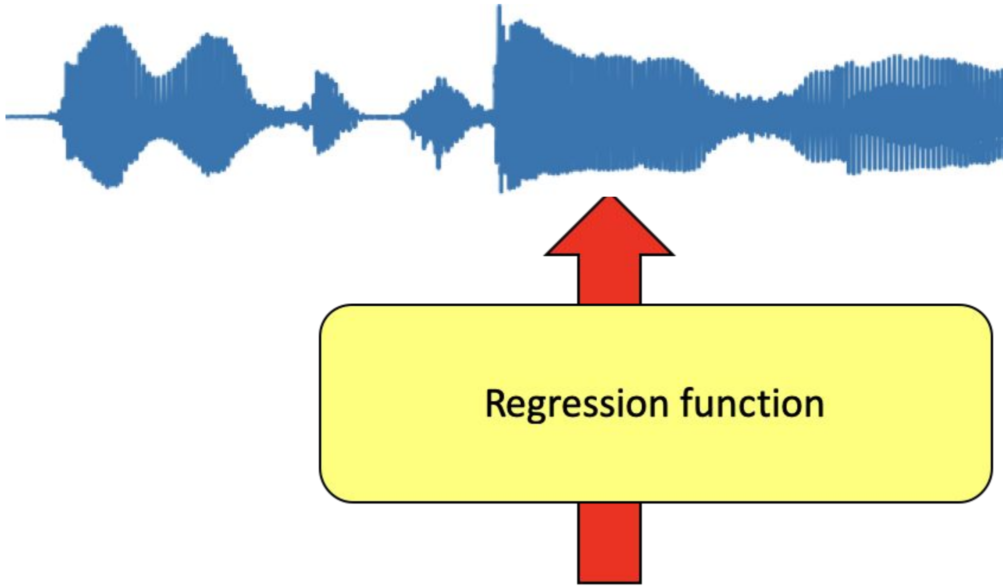
- Basic introduction to Speech Synthesis
- Components of TTS
- Traditional Methods
- Advanced TTS Systems
- Developing TTS Systems
- Future Directions

Definitions

- Speech synthesis is the artificial production of human speech (Wikipedia)
- A text-to-speech (TTS) system converts input text into artificial speech
- In TTS, text analysis and waveform generations are the two important modules
- The ideal requirements of speech synthesizer:
    - Intelligibility: Message content
    - Naturalness: Similarity with the human speech
    - Expressive speech

Text-to-Speech

- An ill-posed problem: Text a narrow band information – to Speech a wide band information



There was a change now

Towards human-like talking machines

For realizing natural human-computer interaction, speech synthesis systems are required to have an ability to generate speech with:

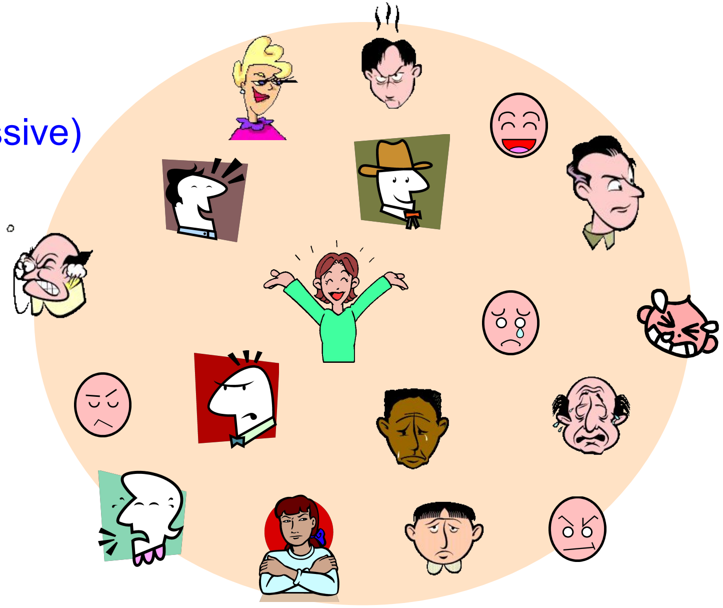
- Arbitrary speaker’s voice
- Speaking styles

(e.g., shouting, read speech, empathy, Dull/expressive)

- Emphasis
- Emotional expressions

  (e.g., happy, angry, sad)

- and so on



Source: HTS web page http://hts.sp.nitech.ac.jp/

Applications

- e-book and email reader
- Alternate Communication for people with disabilities
- Spoken dialogue system: Voice bot, Voice assistant
- Speech-to-speech Translation
- Communication robots

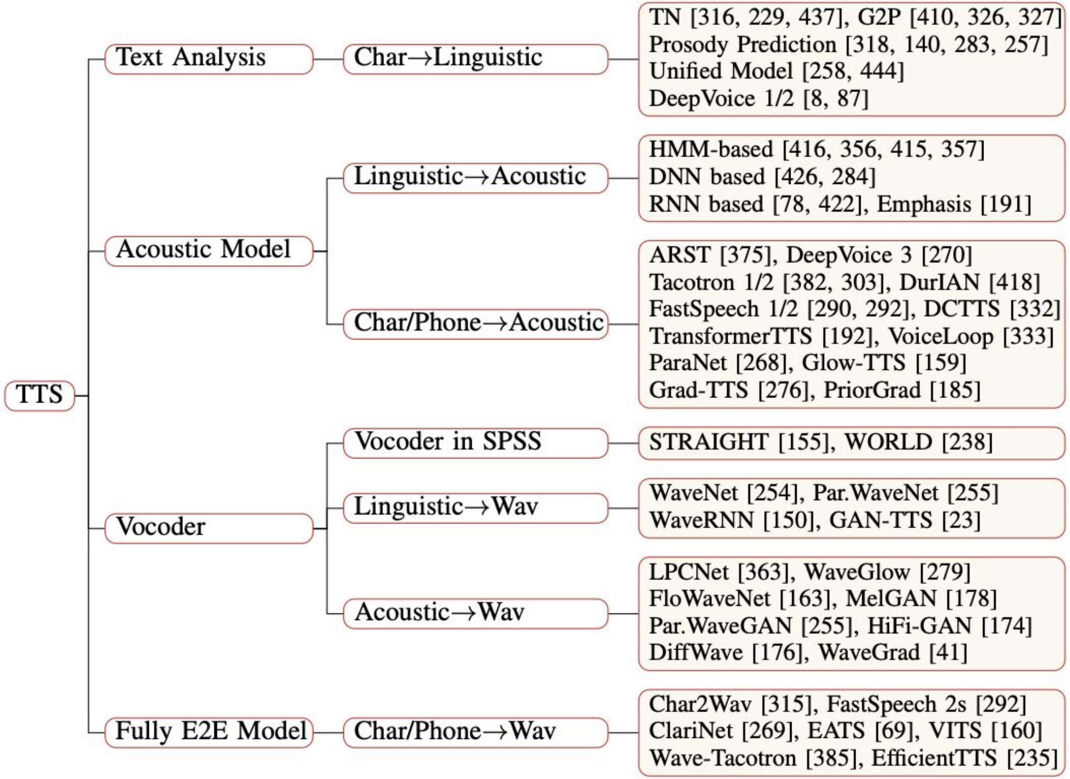
Evaluation Criteria

- Subjective Metric: Evaluated using mean opinion score
  - Naturalness:
    - Rating 1-5
    - Acceptable: ≥4.0
    - Production ready: ≥4.3
  - Speaker Similarity:
    - Rating: 1-5
    - Acceptable: ≥3.8
    - Production ready: ≥4.2
- Objective Metric:
  - Word Error Rate:
    - Range: 0-100%
    - Industry standard: ≤5%
    - High-quality systems: ≤2%
  - PESQ (Perceptual Evaluation of Speech Quality)
  - Mel Cepstral Distortion (MCD)
  - F0 RMSE (Fundamental Frequency Root Mean Square Error)

Industry Requirements

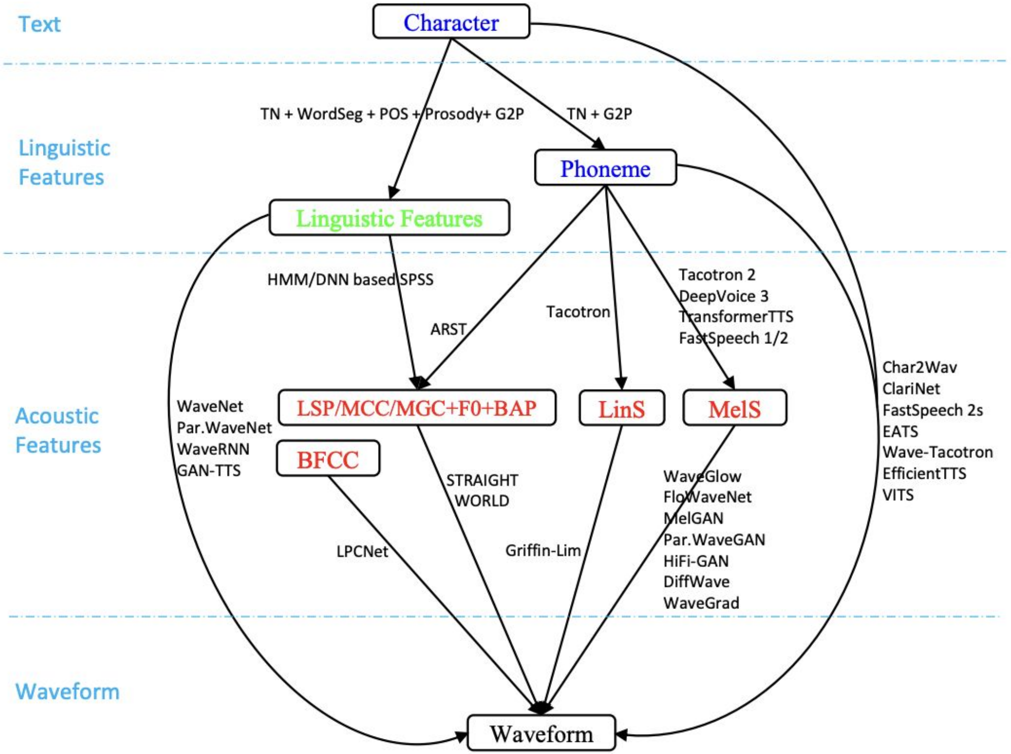
- Correct Pronunciation
- High quality audio
- Low latency (< 100 ms)
- Language/accent support
- Scalable Model

Components in TTS



X. Tan, T. Qin, F. Soong, and T.-Y. Liu, “A survey on neural speech synthesis,” arXiv preprint arXiv:2106.15561, 2021.

Common Features



(b) The data flows from text to waveform.

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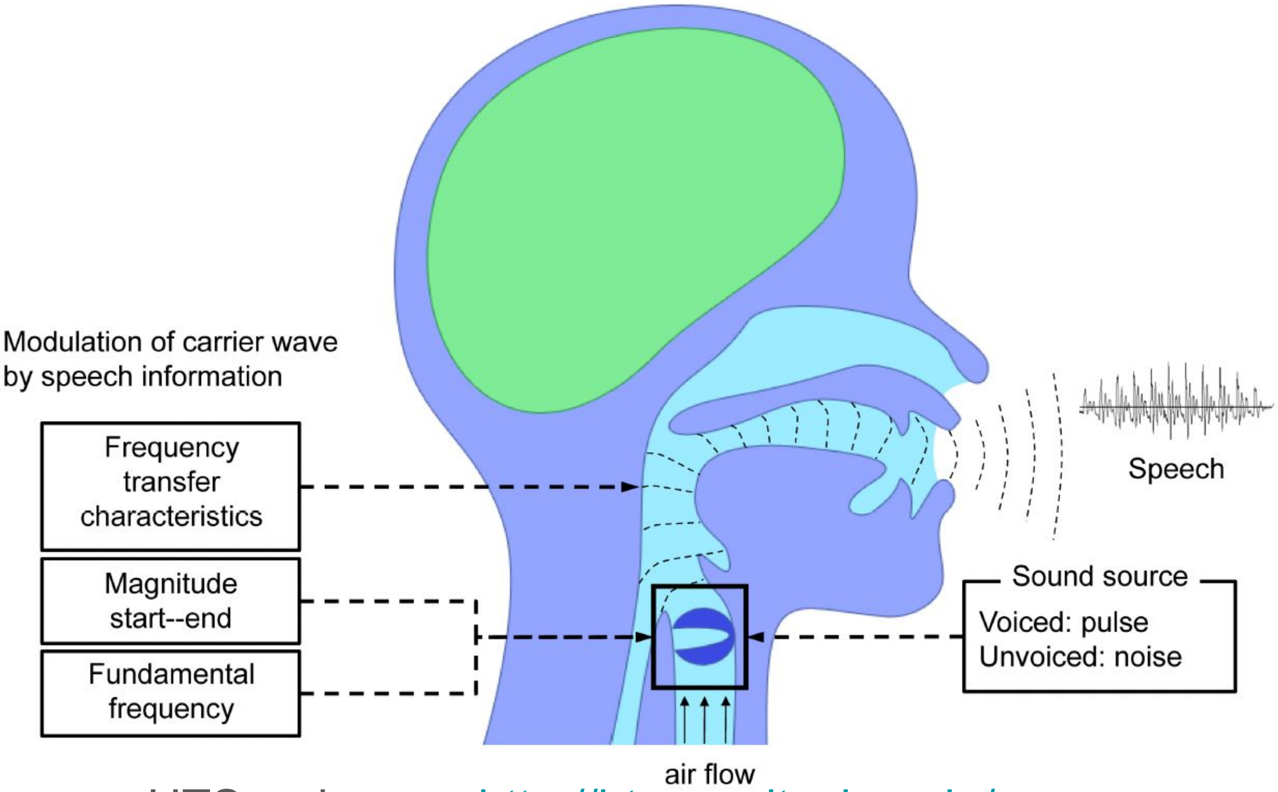
Speech Production Model



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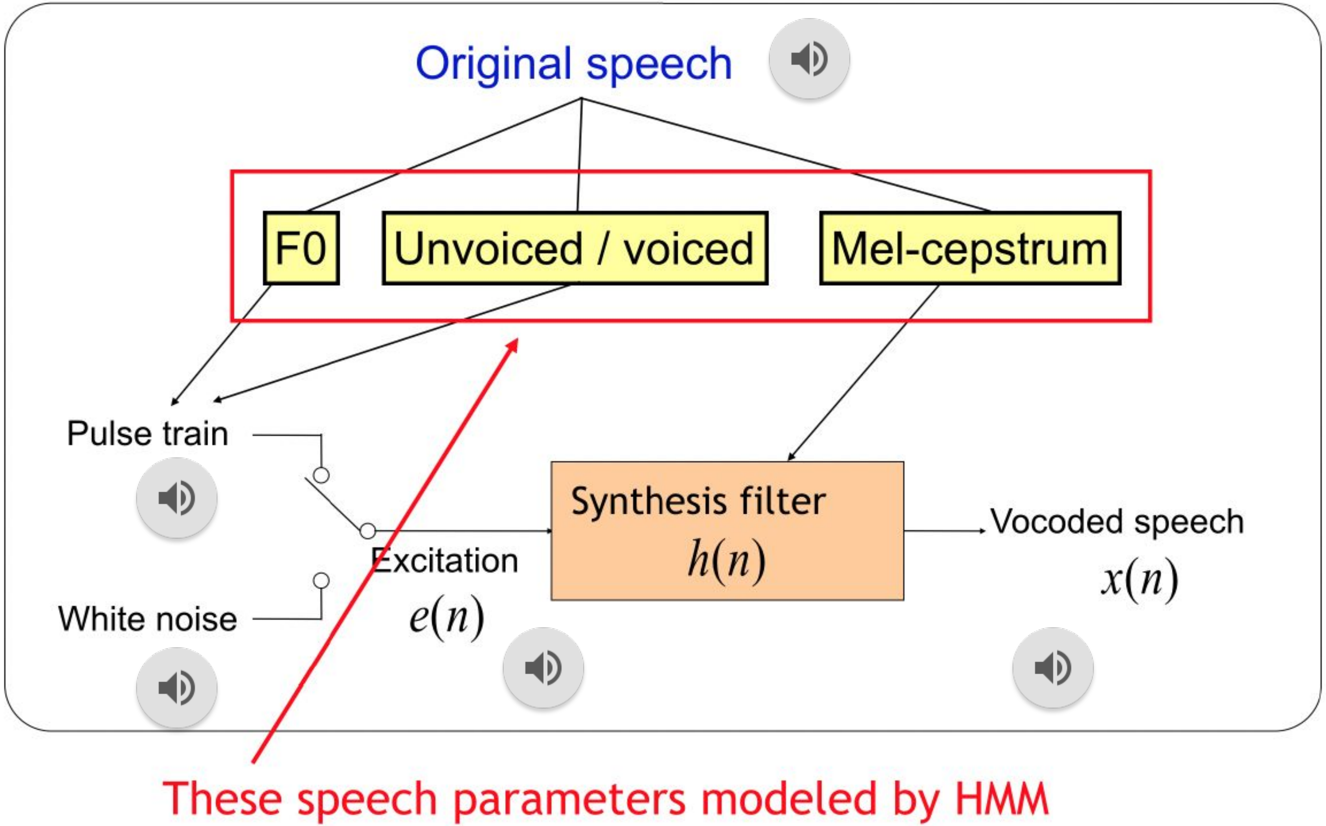
Source filter model



Methods

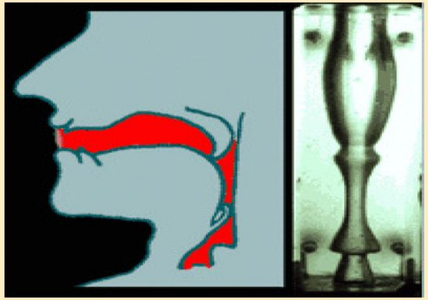
- Formant Speech Synthesis
- Concatenative Speech Synthesis
- Statistical Parametric Speech Synthesis
    - HMM based Speech Synthesis
    - DNN Based Speech Synthesis
- Advanced Speech synthesis
    - Deep neural networks
    - Architectures like flow/Diffusion based models
    - LLM Based models

Formant Speech Synthesis

- It models vocal tract resonances
- Generates speech by controlling formant frequencies and uses source-filter model (physical modelling synthesis).
- Parameters such as fundamental frequency, voicing, and noise levels are varied over time to create a waveform of artificial speech. This method is sometimes called *rules-based synthesis*
- Formant-synthesized speech can be reliably intelligible with less computation.
- However, Quality of speech feel like robotic

Approximation of Vocal tract shape

AH:



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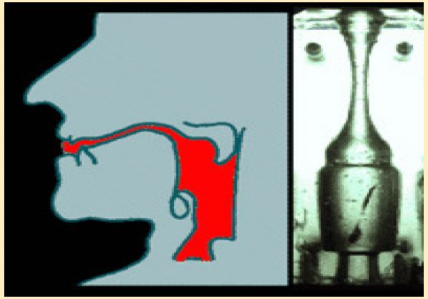

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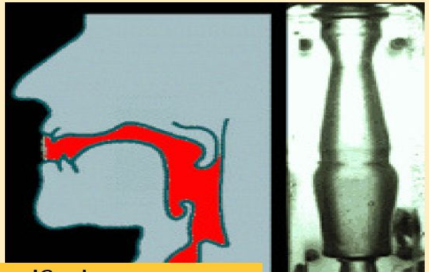

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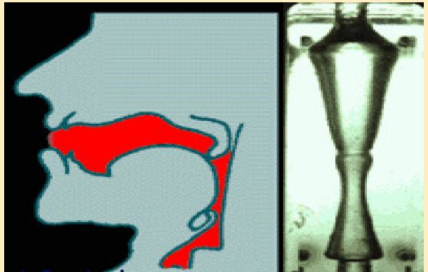


Synthesis from Traditional method

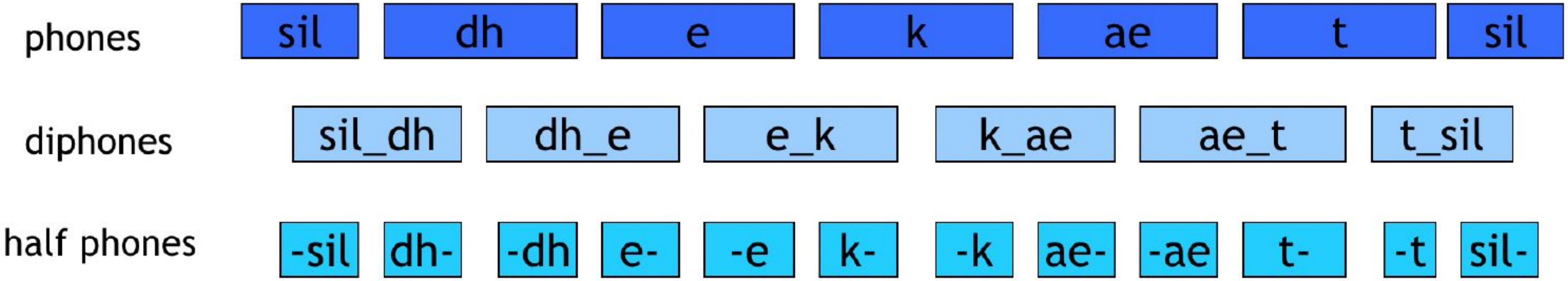
Figure 2: SIGSALY - encrypted voice communications system used in WWII, built in 1943 by Bell Labs engineers (image: Wikipedia)

Concatenative Based Speech Synthesis

- Create speech through concatenation, focus on naturalness.
- Concatenative units are taken from the database of original speech recordings.
- There are multiple methods:
    - Unit selection speech synthesis system.
        - Large database of units with multiple example.
        - Optimal units are selected by searching through the database.

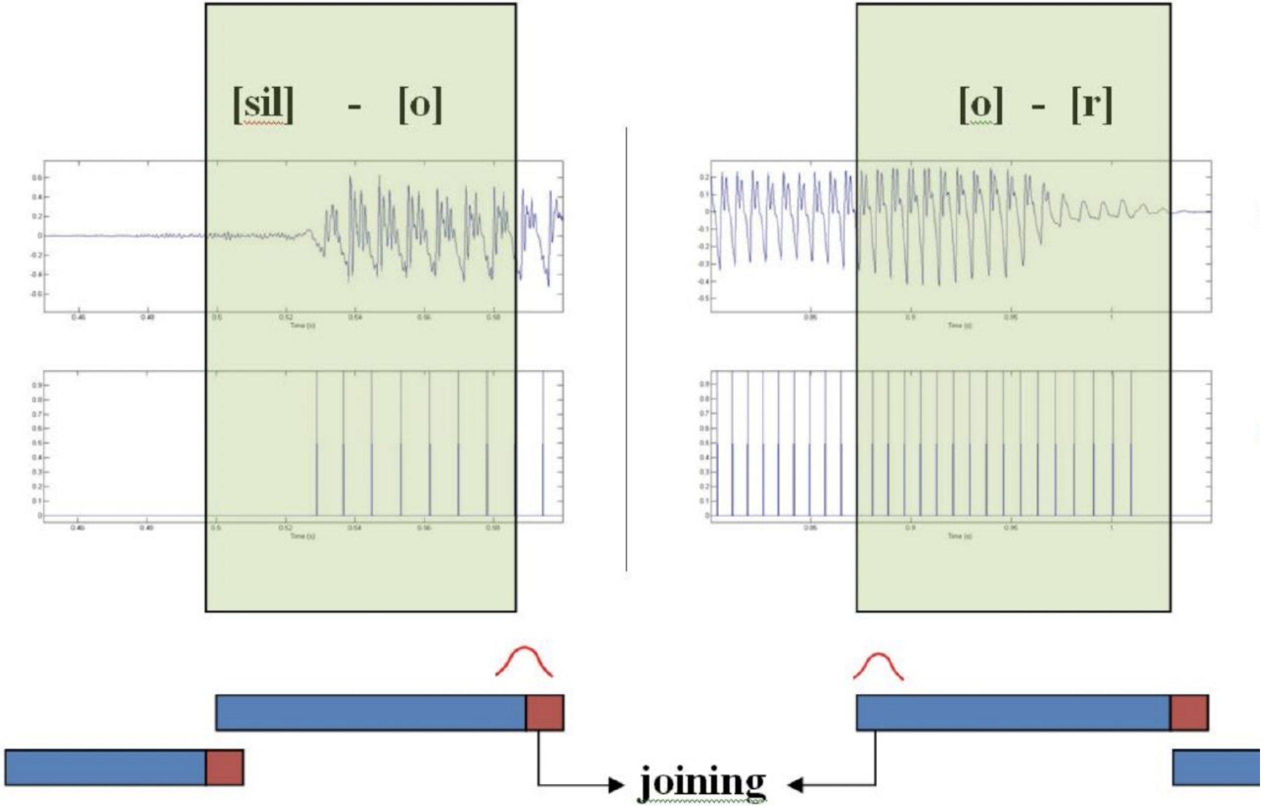
Types of Units

- Diphones, phones, half phones and syllable.



- Diphones provide less spectral distortion at the concatenation points.
- To get the phonetic coverage large database is required.

Concatenation of selected units



Sample

References

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- K. Tokuda, Y. Nankaku, T. Toda, H. Zen, J. Yamagishi, and K. Oura, “Speech synthesis based on hidden markov models", Proceedings of the IEEE, vol. 101-5, pp. 12341252, 2013.
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- Tokuda, K., Yoshimura, T., Masuko, T., Kobayashi, T., Kitamura, T. Speech parameter generation algorithms for HMM-based speech synthesis" in proceedings of ICASSP 2000.

References

- ISCA Synthesis Interest Group: https://synsig.org/index.php/Main_Page (look there for the SPCC videos)
- CSTR, Simon Kings' page on speech synthesis: http://www.speech.zone/courses/speech-synthesis/
- CMU Festvox: http://festvox.org/
- HTS web page http://hts.sp.nitech.ac.jp/ for more information on HMM based speech synthesis system (HTS)
- Google: https://google.github.io/tacotron/
- Espnet toolkit link: https://github.com/espnet/espnet

Thank You